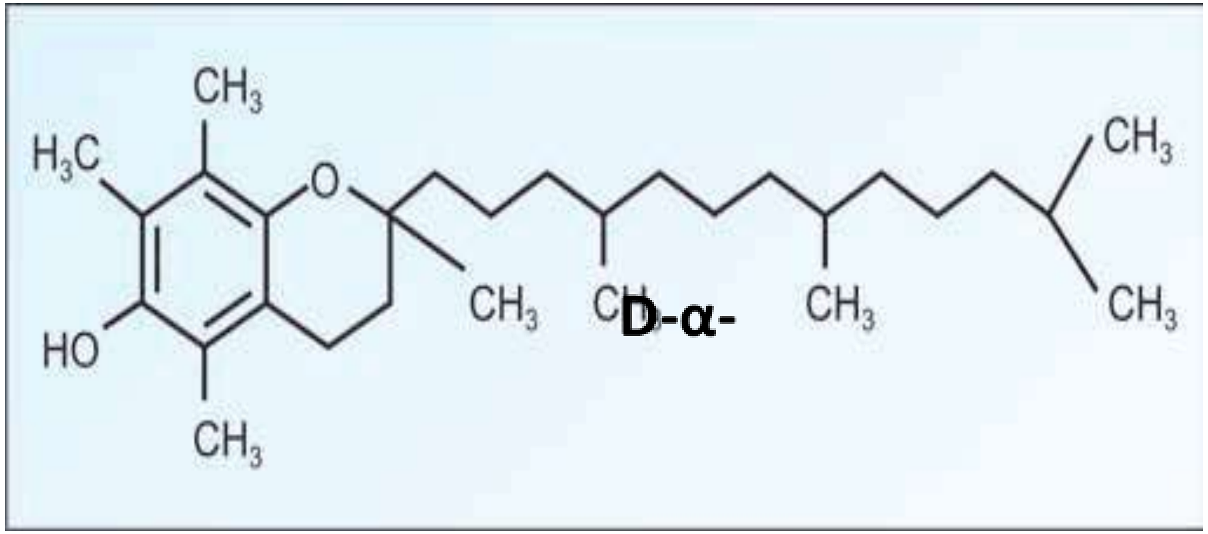
Vitamin 2025

Aml Almakhzoum –part 2

Vitamin E (Tocopherols)



1. There are four types of tocopherols α, B , y and *6.*
2. All contain **tocol** ring.
3. The most active member is **D-*α*- tocopherol**.
4. All of these type of tocopherols different in number and position of - CH3 groups attached to the tocol ring

- **Sources of Vitamin E**

present in **high** amount in **plants** (Wheat germ, Nuts, Strawberries, Vegetable and seed oils), but Liver, and   Eggs yolks contain moderate amounts.

- Its stable to heat and acids
- Cooking in lipids destroy 70-90% of vit E activity
- Storage (freezing) lower the activity
- **RDA** (15IU)

Function

- **Vitamin E Is the Major Antioxidant** in Cell Membranes & Plasma Lipoproteins.

- **The main function** of vitamin E is as **a chain-breaking, free-radical-trapping antioxidant** in cell membranes and plasma lipoproteins, by reacting with the lipid peroxide radicals formed by peroxidation of polyunsaturated fatty acids i.e

  Protects the cell membrane (RBCs, Lung, Heart) and plasma lipoproteins)

- Prevents the alteration of **cell's DNA** and risk of cancer development

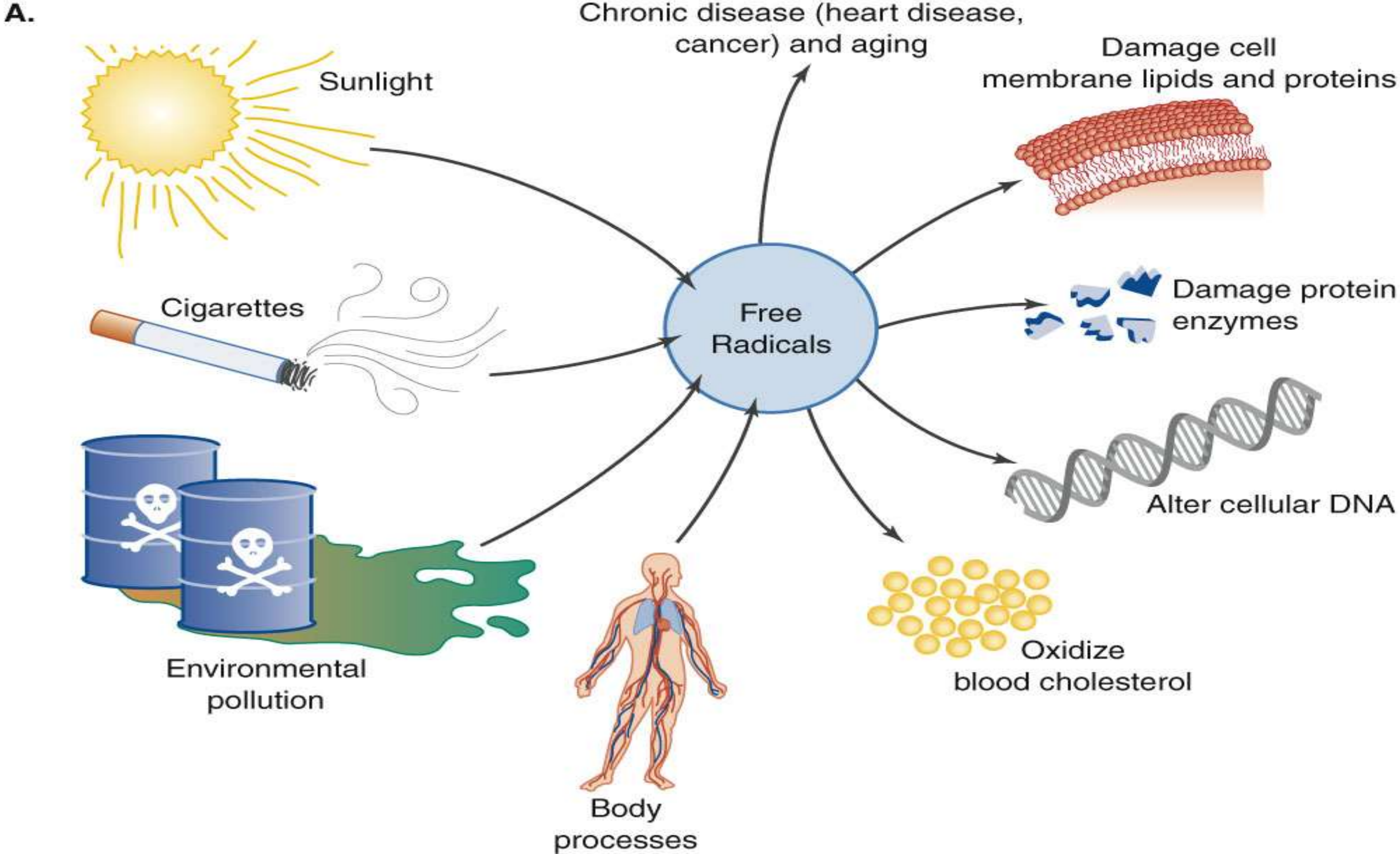
function

- Plays an important role in the development and function of the immune system (protect WBCs)
- Important for reproduction
- Slow biological aging
- Necessary for normal nerve development

- **Free radicals** may lead to chronic disease and tissue damage. (heart disease, cancer, aging)
- Free radicals are produced by body metabolism (nitrous oxide NO), sun light, alcohol, cigarettes, environmental pollution, stress
- Examples of vitamins that contain antioxidants are… **C, E, and Beta Carotene**.

Free Radicals and Anti-Oxiadnts

unstable oxygen molecules can be formed from sunlight, smoking, and pollution



Deficiency of vitamin E

Occurs In cases of severe fat malabsorption, liver disease.

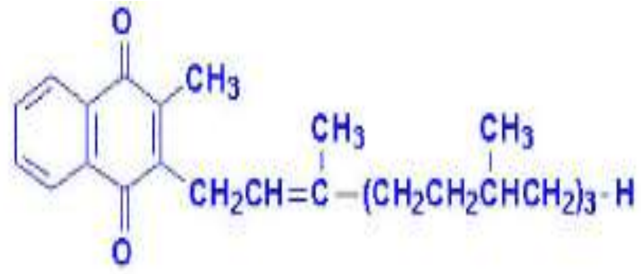
leads to Anemia, Neurological disorders, senile cataract, sterility, andvmuscular dystrophy.

premature infants:- Are born with inadequate reserves of the vitamin E, this may lead to hemolytic anemia.

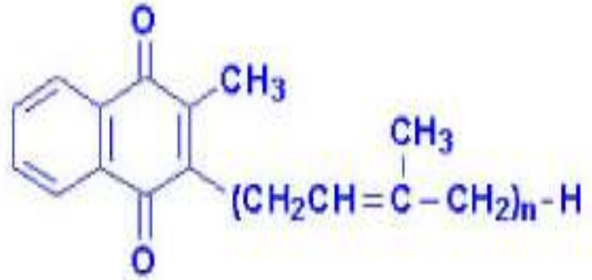
- **Toxicity of vitamin E**
- Vitamin E is the **least toxic of** the fat-soluble vitamins.
- At doses above 1000 IU per day, it may cause

tendency to hemorrhage

Vitamin K: antihaemorrhagic vitamin



Vitamin K1



Vitamin K2
"n" can be 6, 7 or 9
isoprenoid groups

Vitamin K3

Three forms(vitamers) have the biological activity of vitamin K, The difference between K 1

and k2 lies In side chain R.

-(K1) phylloquinone, Found mainly in oil seeds, Green leafy vegetables and tomatoes, nuts, fruits.

-(K2) menaquinone,, synthesized by intestinal bacteria (large intestine) and found in animals

-(K3) Menadione, synthetic Compound that can be metabolized to phylloquinone(more potent than K1,K2)

-RDA: 90 µg/day for adults

Functions of vitamin K

1-Vitamin K is necessary for BLood coagulation.

- Factors dependent on vitamin K are Factor **II** (prothrombin); Factor **VII,** Factor **IX**, and Factor **X**

- All these factors are synthesized by the liver as inactive **zymogens**. They undergo post-translational modification; gamma **carboxylation of glutamic acid** residues. These are the binding sites for calcium ions.

- Vit K act as a **cofactor for the carboxylation** of glutamic acid residues present in these proteins.

- Example:
- **Preprothrombin (**has glutamate residues) by carboxylation convert to →

Prothrombin (has ɣ-carboxy glutamate) →convert to Thrombin has <u>calcium- carboxy glutamate </u>which activates platelets

2-Biosynthesis of other calcium-binding proteins (Osteocalcin) :

- It is calcium binding protein present in bones.
- It is important for proper mineralization of bones.
- 1,25(0H)203 stimulates its synthesis in the form of **pro-osteocalcin.**
- Vitamin K converts pro-osteocalcin into active osteocalcin (by mechanism similar to that of prothrombin i.e. carboxylation of glutamic acid residues).

-Osteocalcin:- also contains hydroxyproline, so its synthesis is dependent on **vitamins K, C, D**

Some drugs such as Dicumarol and Warfarin are **analogs of vit K**, they act as competitive inhibitors with vitamin K for its site of action.

- They are used in treatment of thrombosis
- Also vitamin K is used as antidote to poisoning by dicumarol

Deficiency:

-vitamin K deficiency is **unusual** because adequate amounts are generally produced by intestinal
-bacteria or obtained from the diet
-Sometimes decrease due to : over use of Antibiotics. And immature or damaged liver.

-Leading to increase bleeding time / hemorrhage

-Newborns are injected with a single intramuscular dose of vitamin K because they are susceptible to hemorrhage (**Sterile intestine**)

Vitamin K Toxicity

- More than (1000 mg/day) of **Menadion (K3)** can cause hemolytic anemia, (rupture of RBCs and jaundice)

  liver and kidney damage

Therefore, **menadion** is no longer used to treat vitamin K deficiency.